

On the invariant theory of forms in n variables

In the projective invariant theory of forms in n variables, the principal problem has been solved: the finiteness of the system of forms has been proved. A number of further questions, however, have not been treated, namely those that concern methods for investigating the relations among forms. I should like briefly to communicate the results that I have obtained with respect to such questions, but for the sake of clarity I shall first sketch the questions in the ternary case, where they are already known.

I think first of the two fundamental theorems of the symbolic method: the theorem that all invariant formations can be represented symbolically, and the second theorem that all relations among invariants are obtained by successive application of a small number of symbolic identities; in other words, that the symbolic method gives all relations without leaving the domain of invariants.

Built upon these are the processes for generating forms: the symbolic contraction process, the non-symbolic differentiation processes, the theory of expansions into series, and so on. In this connection I should also mention a theorem proved in my dissertation: all contractions can be generated by successive application of the two possible *cogredient* contractions; by this I mean, for a symbolic product s, t, u, v , the two contractions (stu) and (vx) . Upon this theorem one can build the theory of reducentia and the reduction of systems of forms, analogously to the binary case, as I showed in my dissertation.

When we pass to the n -ary case, already the first theorem of the symbolic method holds only in a qualified sense. As is well known, for quaternary forms there already appear, besides point and plane coordinates x and u , line coordinates p_ρ , namely the minors obtained from two rows of point coordinates. Analogously, in the n -ary case we have rows of variables p_ρ whose individual elements are the minors taken from ρ rows x , and symbolic rows S_ρ whose elements behave like minors taken from ρ rows s , the latter being cogredient to the u .

The symbolic representation by determinants is known to hold only when the symbolic rows S_ρ are resolved into the rows s and these are distributed among different determinants; then only those aggregates of determinants have a real meaning that depend on the S_ρ . Which determinants depend only on the S_ρ can, to be sure, be recognized by differential equations; but in that way one obtains no survey of the totality of invariant formations, nor of the processes by which they are generated.

At this point the theorem on corresponding matrices becomes decisive. To every matrix product

$$(v^{(1)} \dots v^{(\rho)} \mid y^{(1)} \dots y^{(\rho)}),$$

where the rows of the second matrix are contragredient to those of the first, there corresponds a determinant; and conversely every determinant can be transformed into a matrix product with a prescribed number ρ of rows. Hence determinant and matrix product appear as completely equivalent, and the first theorem of the symbolic method takes the following form:

All invariant formations can be represented symbolically by matrix products.

From two symbolic rows S_ρ, T_τ one can now, with the help of rows of variables, form very easily a matrix product that contains these symbolic rows explicitly and therefore represents an invariant formation of the type $(S_\rho | p_\rho)(T_\tau | p_\tau)$, namely

$$(G_{n-\rho-\lambda} S_\rho T_\tau | p_{\rho+\lambda} p_{\tau-\lambda}), \quad \lambda = 1, 2, \dots, \tau \text{ or } n - \rho. \quad (1)$$

More important is the converse statement: all invariant formations can be represented explicitly by matrix products, so that the above process is the genuine extension of the binary and ternary contraction process. To prove this converse it is necessary to transfer the second theorem of the symbolic method to the n -ary case, namely to determine the totality of the independent indecomposable identities in which all rows S_ρ, p_ρ are treated as indecomposable. These identities arise from the known identities containing only rows x and u by replacing the product theorem for determinants by a system of product theorems for matrices and then combining several matrix products into a single one by means of these same product theorems. In this way one arrives at two dual formulas, of which I cite only one:

$$\sum_{i=1}^k (S_{\rho_1} \cdots S_{\rho_k} | p_\rho x^{(i)})(S_\rho | x^{(i)}) = (S_{\rho_1} \cdots S_{\rho_k} S_\rho | p_\rho). \quad (2)$$

The only specialization contained in these formulas is that a row x appears. For completely general symbolic and variable rows one reaches a *decomposition identity*, that is, an identity in which a row p_ρ appears as though resolved into the individual rows x . Clebsch already used the simplest special case of the decomposition identity in his “fundamental problem of invariant theory” in order to derive the elementary covariants occurring in series expansions.

With the aid of these identities, which mediate the passage from the non-explicit determinant expressions to matrix products, one can in fact prove that matrix representation yields the desired explicit symbolic representation.

The same formalism then gives the differentiation processes — for a row of variables gives rise to a row of symbols by replacing its entries with differential symbols — and by means of the decomposition identity it becomes possible to transfer Mertens’s proof almost literally to the n -ary case.